

```
plot(df[["Conc"]],ols_fit.predict(),color="k",zorder=3)
figure(1)
plot(x,y,"o",markerfacecolor="lightgreen",zorder=3)
plot(df[["Conc"]],wls_fit.predict(),color="k",zorder=3)
```

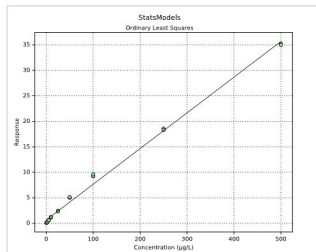


Figure 5: StatsModels OLS

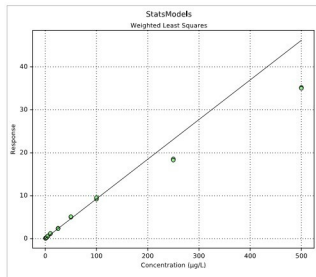


Figure 6: StatsModels WLS

Even for the regression models, the Python language offers different ways to calculate the same things. Some of them have been presented here. I would like to say something about the piecewise, or segmented, regression models. In Python there are, for example, some techniques based on *numpy.piecewise* and *scipy.optimize*, but I'm not completely satisfied by them. For a segmented regression, as previously discussed in the article, 'Get analytical with R' published in September 2015 issue of *OSFY*, I prefer to use R with the package *segmented*. An interesting Python port called *pysegmented* is available from <https://github.com/e-koch/pysegmented> but, at present, it's only a partial

| OLS Regression Results |                  |                     |          |       |                    |
|------------------------|------------------|---------------------|----------|-------|--------------------|
| Dep. Variable:         | Resp             | R-squared:          | 0.999    |       |                    |
| Model:                 | OLS              | Adj. R-squared:     | 0.999    |       |                    |
| Method:                | Least Squares    | F-statistic:        | 2779.27  |       |                    |
| Date:                  | Sun, 14 Aug 2016 | Prob (F-statistic): | 2.79e-27 |       |                    |
| Time:                  | 16:11:23         | Log-Likelihood:     | -22.401  |       |                    |
| No. Observations:      | 20               | AIC:                | 48.96    |       |                    |
| DF Residuals:          | 18               | BIC:                | 58.95    |       |                    |
| DF Model:              | 1                |                     |          |       |                    |
| Covariance Type:       | nonrobust        |                     |          |       |                    |
|                        |                  |                     |          |       |                    |
|                        | coef             | std err             | t        | P> t  | [95.0% Conf. Int.] |
| Intercept              | 0.0404           | 0.206               | 3.154    | 0.005 | 0.217 1.002        |
| Conc                   | 0.0761           | 0.001               | 61.403   | 0.000 | 0.068 0.072        |
|                        |                  |                     |          |       |                    |
| Omnibus:               | 0.066            | Durbin-Watson:      | 0.382    |       |                    |
| Prob(Omnibus):         | 0.819            | Jarque-Bera (JB):   | 0.920    |       |                    |
| Skew:                  | 1.101            | Prob(SB):           | 0.058    |       |                    |
| Kurtosis:              | 3.276            | Cond. No.           | 212.     |       |                    |

Warnings:  
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 7: OLS summary

| WLS Regression Results |                  |                     |         |       |                    |
|------------------------|------------------|---------------------|---------|-------|--------------------|
| Dep. Variable:         | Resp             | R-squared:          | 0.978   |       |                    |
| Model:                 | WLS              | Adj. R-squared:     | 0.968   |       |                    |
| Method:                | Least Squares    | F-statistic:        | 571.5   |       |                    |
| Date:                  | Sun, 14 Aug 2016 | Prob (F-statistic): | 3.4e-15 |       |                    |
| Time:                  | 16:11:23         | Log-Likelihood:     | 1.9341  |       |                    |
| No. Observations:      | 20               | AIC:                | 6.2118  |       |                    |
| DF Residuals:          | 18               | BIC:                | 2.123   |       |                    |
| DF Model:              | 1                |                     |         |       |                    |
| Covariance Type:       | nonrobust        |                     |         |       |                    |
|                        |                  |                     |         |       |                    |
|                        | coef             | std err             | t       | P> t  | [95.0% Conf. Int.] |
| Intercept              | 0.0395           | 0.005               | 7.422   | 0.000 | 0.028 0.051        |
| Conc                   | 0.0923           | 0.004               | 24.031  | 0.000 | 0.084 0.100        |
|                        |                  |                     |         |       |                    |
| Omnibus:               | 0.209            | Durbin-Watson:      | 0.605   |       |                    |
| Prob(Omnibus):         | 0.901            | Jarque-Bera (JB):   | 0.409   |       |                    |
| Skew:                  | 0.877            | Prob(SB):           | 0.815   |       |                    |
| Kurtosis:              | 2.316            | Cond. No.           | 1.96    |       |                    |

Warnings:  
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 8: WLS summary

port. More complex applications can be done with the Scikit-learn package (Reference 9), but this is another story, perhaps for another article in the future. [END](#)

## References

- [1] Lantz, *Machine learning with R* Packt Publishing, Birmingham, 2013.
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- [3] Meier, Zünd, *Statistical methods in analytical chemistry*, Wiley, New York, 2000.
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- [5] Cavalli, Sarzanini, *La cromatografia ionica nell'analisi ambientale* (The ion chromatography in the environmental analysis), Nuova Poligrafica Milanese, Milan, 1999.
- [6] Kjeldahl, Bro, *Common misunderstandings in chemometrics*, Journal of Chemometrics, 2010, 24, 558-564.
- [7] <http://people.stfx.ca/bliengme/exceltips/regressionanalysisconfidence2.htm>, last visited on 12/08/2016.
- [8] <http://statsmodels.sourceforge.net/devel/examples/index.html>, last visited on 12/08/2016.
- [9] [http://scikit-learn.org/stable/supervised\\_learning.html](http://scikit-learn.org/stable/supervised_learning.html), last visited on 12/08/2016.

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